

Federated Study Readiness Checklist

For medical AI projects where patient-level data may not move

Use before the first federation meeting

A project is not ready because the algorithm has a name. It is ready when the clinical question, data movement, harmonization, evaluation, privacy, and governance are explicit.

1 Clinical question

- ☐ Target population, intended use, and decision context are explicit.
- ☐ Outcome is clinically meaningful and measured similarly across sites.
- ☐ A non-FL baseline and site-held-out validation plan are defined.

2 Data movement

- ☐ Patient-level centralization was evaluated and documented.
- ☐ Permitted outputs are specified: updates, metrics, aggregates, logs.
- ☐ Disclosure rules are written for counts and reporting.

3 Harmonization

- ☐ Cohort definitions and exclusions are site-tested.
- ☐ Data dictionary, label protocol, and preprocessing are versioned.
- ☐ Missingness, timing, and annotation quality are profiled by site.

4 Evaluation

- ☐ Per-site, worst-site, subgroup, and global metrics are specified.
- ☐ Calibration and clinical utility are included, not only discrimination.
- ☐ Failure criteria and stop rules are defined.

5 Privacy and security

- ☐ Threat model covers honest-but-curious, compromised, and malicious participants.
- ☐ Secure transport, authentication, authorization, and audit logging are assigned.
- ☐ Update review, poisoning detection, and incident response are included.

6 Operations and governance

- ☐ Each site has clinical, technical, and governance owners.
- ☐ Ethics, data-protection, contracts, authorship, and model ownership are documented.
- ☐ Maintenance, withdrawal, deletion, and publication plans are agreed.

Red flag: Pause the FL plan if the governance problem is being handed to the algorithm, or if the only privacy statement is that data stay local.